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Prelab: 7

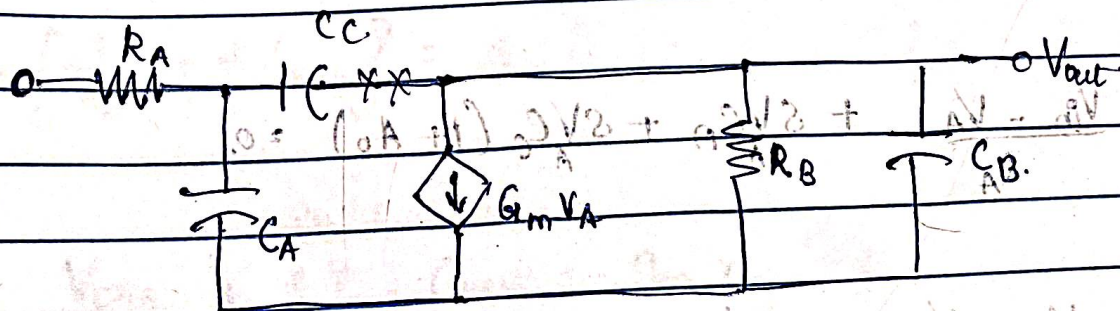
- 1) Create a table ranking the various amplifiers as good, medium or poor in the following categories: gain, input impedance, output impedance and linearity. Include the expression

Amplifier Type	Voltage Gain	Input Impedance	Output Impedance	Linearity
	$A_v = -G_m R_{out}$			
1) Current Mirror Load	$A_v = -g_{m1}(r_{o2} \parallel r_{o6})$	$R_{in} = 1/g_{m1}$ (Medium)	$R_{out} = r_{o2} \parallel r_{o6}$ (High)	(Poor)
2) Digital CMOS Inverter	$A_v = (-g_{m1} + g_{m2})(r_{o1} \parallel r_{o2})$ High (G)	$R_{in} = \infty$ (MOS gate driven) Good.	$R_{out} = r_{o1} \parallel r_{o2}$ (Medium)	Poor (P)
3) Self Biased P-MOS Inverter	$A_v = -\frac{g_{m1}}{g_{m2}}$ (Low)	$R_{in} = 1/g_{m1}$ (Medium)	$R_{out} = 1/g_{m2}$ (Low)	Good (G)
4) Self Biased C-MOS Inverter	$A_v = -\frac{g_{m1}}{g_{m2}} \rightarrow$ (Low)	$R_{in} = 1/g_{m1}$ (Medium)	$R_{out} = 1/g_{m2}$ (Low)	Medium Good.

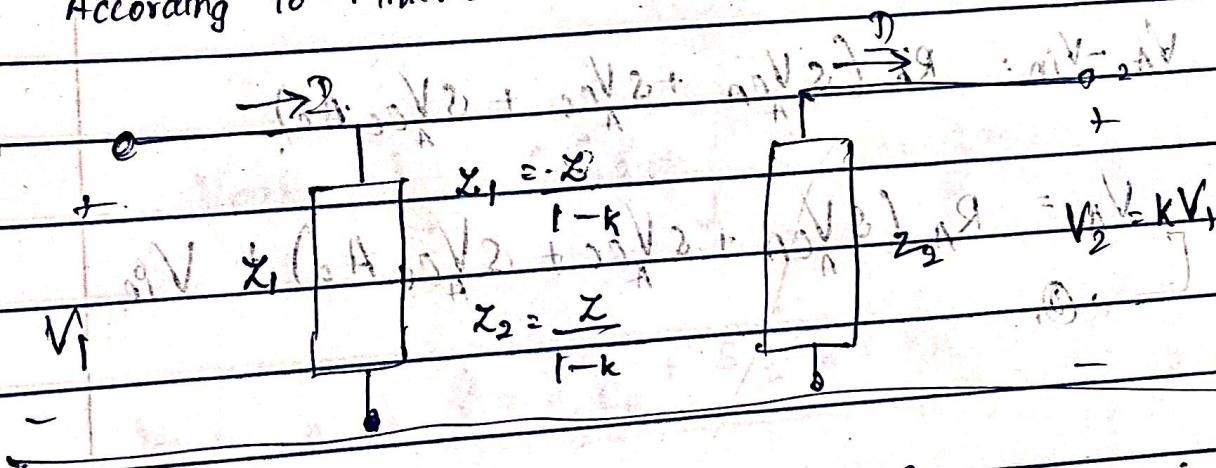
2) Derive the transfer function for the generic amplifier by applying Miller's theorem. Compare this transfer function to the one derived in the manual. Comment on the utility of Miller's theorem.

input capacitance : C_m

output capacitance : C_{out}



According to Miller's theorem:



Converting the circuit into Miller's form,

$$C_{in} = C_A + C_C (1 + A_0)$$

$$C_{out} = C_B + C_C \left(1 + \frac{1}{A_0} \right)$$

Applying KCL @ ~~2~~

$$\frac{V_{in} - V_A}{R_A} + s V_A C_{in} = 0$$

$$V_{in} - V_A + s R_A V_A C_{in} = 0$$

$$V_A (s R_A C_{in} - 1) = -V_{in}$$

$$V_{in} = (1 - s R_A C_{in}) V_A$$

Applying KCL on 2nd loop,

$$\frac{V_{out}}{R_B} + s V_{out} C_{out} = [g_m V_A + \frac{1}{s R_A}] V_{in}$$

$$V_{out} \left[\frac{1}{R_B} + s C_{out} \right] = \frac{g_m V_{in}}{(1 - s R_A C_{in})}$$

$$V_{out} \left[1 + s R_B C_{out} \right] = \frac{g_m V_{in}}{(1 - s R_A C_{in})}$$

$$\frac{V_{out}}{V_{in}} = \frac{g_m R_B}{(1 + s R_B C_{out}) (1 - s R_A C_{in})}$$

$$\frac{V_{out}}{V_{in}} = \frac{g_m R_B}{(1 + s R_B C_{out}) (1 - s R_A C_{in})}$$

$$\frac{V_{out}}{V_{in}} = \frac{g_m R_B}{\left(1 + s R_B \left(C_B + C_C \left(1 + \frac{1}{A_o}\right)\right)\right) \cdot \left(1 + s R_A \left(C_A + C_C \left(1 + A_o\right)\right)\right)}$$

$$= \frac{g_m R_B}{\left(1 + s R_B C_B + s R_B C_C + s R_B \frac{C_C}{A_o}\right) \left(1 - s R_A C_A + s R_A C_C + s R_A C_C A_o\right)}$$

$$\left(1 + s R_B C_B + s R_B C_C + s R_B \frac{C_C}{A_o}\right) \left(1 - s R_A C_A + s R_A C_C + s R_A C_C A_o\right)$$

Now comparing both,

According to Manual,

$$H(s) = \frac{g_m R_B \left(1 - s \frac{C_C}{g_m}\right)}{1 + s R_B (C_A + C_C) + s^2 R_A R_B C_B C_C}$$

$$1 + s R_B (C_A + C_C) + s^2 R_A R_B C_B C_C$$

3) Design the following inverting Amplifier

100V
10V

(a) Current Mirror load Inverter

$$V_{DD} = -V_{SS} = 0.9V$$

$$R_S = 100k\Omega$$

$$C_L = 1pF$$

20mV

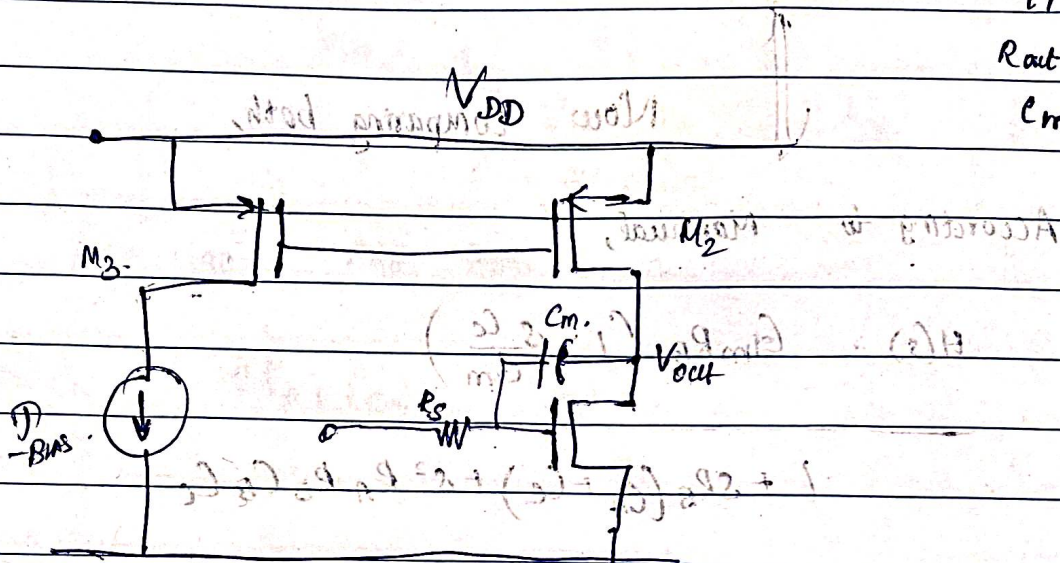
$$GBW = 2MHz$$

$$A_{V_0} = 30dB \quad \text{phase shift @ } U_{GF} < 120^\circ$$

f_{p1}

R_{out}

C_m



Step 1: Determine the Miller compensation capacitor C_m from the gain-bandwidth product (GBW)

$$GBW = A_{V_0} f_{p1}$$

$$\text{also, } A_{V_0} = -G_m R_{out}$$

$$\frac{A_0}{-G_m} = R_{out}$$

$$\frac{2MHz}{30dB} = f_{p1}$$

$$R_{out} = 20dB$$

$$\frac{2MHz}{10^{80/20}} = f_{p1}$$

$$f_{p1} = 0.0632 MHz$$

$$= 63.245 kHz$$

(b) When the amplifier is connected in feedback, the phase shift measured at the UGF

$$f_{p2} > \sqrt{3} \text{ GBW.}$$

$$g_m = \frac{1}{\lambda \frac{V}{I_{bias}}}$$

$$f_{p2} > \sqrt{3} \times 2 \text{ MHz.}$$

$$f_{p2} > 3.46 \text{ MHz.}$$

$$f_{p2} = \frac{1}{2\pi RC}$$

$$R = R_{out}$$

$$C = C_L + (1 + A_{V_o}) C_m$$

$$\text{Let } f_{p2} = 3.5 \text{ MHz}$$

$$C = 1 \times 10^{-12} + (1 + 10^{30/20}) C_m$$

$$3.5 \text{ MHz} = \frac{1}{2\pi RC}$$

Computing, Miller Capacitance.

$$3.5 \times 10^6 = \frac{1}{2\pi R_{out} C}$$

$$2\pi f_p = \frac{1}{R_{out} C}$$

$$10^{30/20} \times 100 \text{ k}\Omega \times C_m$$

$$R_{out} = \frac{1}{2\pi \times 3.5 \times 10^6 \times C}$$

$$2\pi \times 3.5 \times 10^6 \times C$$

$$C_m = \frac{1}{10^{30/20} \times 100 \text{ k}\Omega \times C}$$

$$= \frac{1}{2\pi \times 3.5 \times 10^6 \times C}$$

$$2\pi \times 3.5 \times 10^6 \times C$$

$$C_m = 0.8 \text{ pF}$$

$$R_{out} = 45.47 \text{ k}\Omega$$

$$C = 10^{-12} + (1 + 10^{30/20}) 0.8 \times 10^{-12}$$

$$f_{p2} = \frac{1}{2\pi R_{out} C}$$

$$= 10^{-12} \left(1 + (1 + 10^{30/20}) 0.8 \right)$$

$$G_m = \frac{1}{10^{30/20} \times (45.47 \text{ k}\Omega)}$$

$$C = 27.01 \text{ pF}$$

$$G_m = 0.0188455$$

$$G_m = 6.95 \times 10^{-5}$$

$$R_{out} g_m = \frac{1}{\lambda P_{bias}}$$

$$R_{out} g_m = \frac{1}{\lambda P_{bias}} \quad \lambda = 158.6 \text{ mV}$$

$$45.47 \frac{P_{K\Omega}}{P_{bias}} = \frac{1}{\lambda P_{bias}}$$

$$P_{bias} = \frac{1}{(0.158.6 \times 45.4 \times 10^3)}$$

$$= \frac{1}{0.01586 \times 695.46 \mu s}$$

$$P_{bias} = 234.57 \mu A \quad P_{bias} = 138.8 \mu A$$

$$P_D = P_{bias}$$

$$\frac{1}{2} k_p$$

$$g_m = \sqrt{2 \mu_n C_{ox} \left(\frac{\omega}{L} \right) P_D}$$

$$0.0188 = \sqrt{2 \times 214.326 \times 844.57 \times \left(\frac{\omega}{L} \right)}$$

$$0.0188 = \sqrt{143414.0996 \left(\frac{\omega}{L} \right)}$$

$$0.00035344 = 143414.0996 \left(\frac{\omega}{L} \right)$$

(b) Digital CMOS Inverter

$$V_{DD} = -V_{SS} = 0.9$$

$$R_S = 100 \text{ k}\Omega$$

$$C_L = 1 \text{ pF}$$

$$\text{GBW} = 2 \text{ MHz}$$

$$A_{V_0} = 30 \text{ dB}$$

$$\text{UGF} < 120^\circ$$

$$\lambda_p = 0.118 \mu\text{V} V_{TN} = 447.026 \text{ mV}$$

$$\lambda_n = 0.158 \mu\text{V} V_{TP} = 421.6376 \text{ mV}$$

$$\textcircled{1} A_{V_0} = -G_m R_E$$

$$A_{V_0} = \frac{-2}{(0.118 + 0.158) \mu} \left(\frac{1}{0.9 - 0.447} + \frac{1}{0.9 - 0.421} \right)$$

$$= 154.0724 \cdot 0 (2.2075 + 2.087)$$

$$A_{V_0} = 31.119$$

$$f_{p1} = \frac{2 \text{ MHz}}{10^{20/20}}$$

Miller Capacitance,

$$2\pi f_{p1} = -1 / (|A_{V_0}| R_S C_m)$$

$$f_{p1} = 64.1 \text{ kHz}$$

$$C_m = \frac{1}{2\pi f_{p1} \times 100 \text{ k}\Omega}$$

$$2\pi \times 64.1 \text{ kHz} \times 100 \text{ k}\Omega$$

$$C_m = \cancel{0.2 \text{ pF}} \quad 24.71 \text{ pF}$$

$$f_{p2} \geq \sqrt{3} \cdot \text{GBW}$$

$$f_{p2} > \sqrt{3} \cdot 2 \text{ MHz}$$

$$f_{p2} \geq 3.46 \text{ MHz}$$

$$K_{P_{N,P}} \left(\frac{W}{L} \right)_{N,P} \geq \frac{g_{m1,2}}{V_{DD} - V_{TP,R}}$$

$$\left(\frac{K_{P_N} \left(\frac{W}{L} \right)_{N,P}}{154.0 - 0.9} \right) \geq \frac{g_{m1}}{V_{DD} - V_{TN} + 0.427}$$

$$\geq \frac{0.0188}{(120.5 + 110.5) \cdot 0.95 + 0.427}$$

$$K_{P_N} \left(\frac{W}{L} \right) \geq 0.0415$$

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$$11.18 \mu\text{A}$$

$$K_{P_{H,S}} = 196$$

$$0.55$$

$$1.43 \text{ A} = 1.43$$

$$(m) \text{ } \frac{1}{1000} \times 1000 = 1$$

$$1000 \times 1000 = 1000000$$

$$1000 \times 1000 \times 1000 = 1000000000$$

$$\left[196 \left(\frac{W}{L} \right) = 196 \cdot 0.21 \right]$$

$$f_{p2} > \sqrt{3} \cdot \text{GBW}$$

$$f_{p2} > \sqrt{3} \cdot 2 \text{ MHz}$$

$$f_{p2} > 3.46 \text{ MHz}$$

$$K_{P,N,P} \left(\frac{W}{L} \right)_{N,P}^2 = \frac{g_{m1,2}}{V_{DD} - V_{TP,R}}$$

$$\left(\frac{K_{P,N}}{154.0 - 1.0} \left(\frac{W}{L} \right)_{N,P}^2 \right) = \frac{g_{m1}}{V_{DD} - V_{TN} + 21.0}$$

$$= \frac{0.0188}{(120.5 + 210.5) \cdot 0.9 - 0.421}$$

$$K_{P,N} \left(\frac{W}{L} \right) = 0.0415$$

$$211.15 \mu\text{A}$$

$$211.15 \mu\text{A} = 196$$

$$(c) A_0 = 201 \quad g_{m1} = -31.09$$

$$g_{m2} = 0.828 \text{ mS}$$

$$L = 270 \text{ n}$$

$$N_{ox} = 621 \mu\text{m}$$

$$(d) \left(\frac{W}{L} \right)_{n,p}^2 = \frac{2(214 \mu\text{A})}{0.9} = 64.9$$

$$\left(\frac{W}{L} \right)_{n,p} = 230$$

$$L = 270$$

$$W = 62.1$$